Kevin **Tongue**

Computational Mechanics & Scientific Machine Learning

Saint-Étienne, France

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Profile

Mechanical engineer with two engineering degrees and a Master's degree in Numerical Solid Mechanics from Centrale Lyon – ENISE, awarded in September 2026. My work combines computational mechanics and scientific machine learning: PDE-based modelling, finite elements, model order reduction (POD, PGD), Physics-Informed Neural Networks and data-driven models built from high-fidelity simulations. Simulation Engineer at EP MECA since October 2026; interested in research at the interface of physics and AI for efficient, physically consistent modelling.

Education

Centrale Lyon – ENISE

Saint-Étienne, France

Engineering degrees in General Mechanical Engineering and Design & Simulation; Master's in Numerical Solid Mechanics

2024 – September 2026

- Numerical methods for PDEs: finite elements, variational formulations, Navier–Stokes methods and nonlinear mechanics.
- Model order reduction (POD, PGD), uncertainty quantification and design of experiments.
- Scientific machine learning: clustering, regression, neural networks, optimal control and Physics-Informed Neural Networks.

National Advanced School of Engineering of Yaoundé (ENSPY)

Yaoundé, Cameroon

Engineering cycle (M1 Mechanics) and B.Sc. in Physical Sciences

2019 – 2024

Mechanical engineering and physical sciences, including materials and manufacturing.

Professional & Research Experience

EP MECA

France

Simulation Engineer

1 October 2026 – Present

- Modelling and numerical simulation of a datacenter cooling system.
- Building the physical model, defining boundary conditions and analysing thermal fields to support design decisions.

IMT Mines Saint-Étienne and IMT Mines Albi

France

Master's Research Intern

9 March – 17 September 2026

- Built a data-driven reduced model of granular segregation by coupling DEM simulations with an inhomogeneous Markov chain; estimated transition matrices from large Parquet simulation datasets.
- Developed an object-oriented Python library for spatial partitioning, clustering-based state definition, post-processing and XDMF/HDF5 export to ParaView; performed Monte Carlo sensitivity analyses.
- **Thesis:** Modelling of granular segregation using inhomogeneous Markov chains (64 pp.) — code: `https://github.com/tiffank1802/DEM-MCM-Coupling`.

SOPECAM

Yaoundé, Cameroon

Engineering Intern – Industrial Diagnostics & Preventive Maintenance

May – September 2024

Diagnosed pump failures using vibration and temperature sensor data; applied FMEA and developed preventive-maintenance software.

Selected Projects

Centrale Lyon – ENISE, collaboration with LTDS

Saint-Étienne, France

Physics-Informed Neural Network for Inverse Material Identification

October 2025 – March 2026

- Built a PyTorch PINN trained on FEM-generated data, embedding heat-conduction and elasticity PDEs in the loss through automatic differentiation to identify constitutive-law parameters.
- Deployed the application on Hugging Face Spaces.

Centrale Lyon – ENISE, team project

Saint-Étienne, France

Fatigue Failure Predictor – Dang Van Criterion

October 2025 – March 2026

Implemented two Python methods for multiaxial fatigue assessment, validated against reference cases and ChronoCad; the geometric method runs approximately 160 times faster than the angular method.

Advanced FEM Modelling practical project

Centrale Lyon – ENISE

Digital Twin of a Soft Robot

October – December 2025

Combined FEM assembly, Newmark time integration, Kalman filtering and PID/LQR control in MATLAB; used real-time data assimilation to reconstruct displacement fields from indirect sensors.

Numerical Mechanics coursework — *Centrale Lyon – ENISE*

Reduced-Order Modelling: POD and PGD — *2025 – 2026*

Implemented snapshot POD with projection and greedy space-time PGD enrichment, including truncation-error analysis.

Publications & Reports

2026 **Tongue K.**, Modelling of granular segregation using inhomogeneous Markov chains (Master's thesis, 64 pp.), read the report — *IMT Mines Albi*

Technical Skills

Scientific ML	PyTorch, PINNs, neural networks, regression, clustering, automatic differentiation
Applied mathematics	PDE modelling, finite elements, Navier–Stokes methods, POD, PGD, uncertainty quantification, Markov chains
Scientific computing	Python (NumPy, SciPy, PyTorch, pandas, Matplotlib), MATLAB/Simulink, C++ (Eigen, deal.II – in progress), Fortran (basics), Linux, Git
Simulation	Abaqus, SolidWorks Simulation, ParaView, HDF5/XDMF data pipelines
Languages	French (native), English (B2, TOEIC)

Certifications

MATLAB (MathWorks) | Google Advanced Data Analytics | IBM Data Analyst